A Comparative Study of Deep RL Algorithms: DQN, PPO, SAC, and TD3 on Four Classic-Control Benchmarks

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Abstract

We re-implement DQN, PPO, SAC and TD3 from scratch in PyTorch and compare them on four Gymnasium classic-control benchmarks with three seeds per pair and 95% bootstrap CIs. Four pre-registered ablations stress one canonical hyperparameter per algorithm. Headline findings: on CartPole DQN is seed-bimodal while PPO is stable; on Acrobot DQN solves cleanly; on dense-reward Pendulum SAC and TD3 are $\sim 13\times$ more sample-efficient than PPO; on sparse-reward MountainCarContinuous only PPO ever finds the goal, and the SAC-α and TD3-σ ablations confirm that the off-policy failure is structural in the canonical per-step exploration mechanism, not a tuning issue.

1 Introduction

This project re-implements four canonical deep RL algorithms from scratch in PyTorch and compares them on four Gymnasium classic-control benchmarks. The algorithms span the two main axes of the deep RL taxonomy: DQN Mnih et al. [2013] is value-based and off-policy, PPO Schulman et al. [2017] is policy-based and on-policy, SAC Haarnoja et al. [2018] is policy-based off-policy with maximum-entropy regularization, and TD3 Fujimoto et al. [2018] is deterministic off-policy actor-critic. Environments are CartPole-v1 and Acrobot-v1 (discrete) and Pendulum-v1 and MountainCarContinuous-v0 (continuous), provided by Gymnasium Towers et al. [2023].

We follow the brief's three asks: (i) a fair empirical comparison with three seeds and bootstrap CIs, (ii) a documented implementation diary of the details that mattered, and (iii) qualitative answers to the four discussion questions on compute, compactness, continuous-action scaling, and off-policy data efficiency. Four empirical headlines: **(1)** on CartPole DQN is bimodal across seeds while PPO is stable; **(2)** on Acrobot DQN solves cleanly in $\sim$30k steps while PPO is slower; **(3)** on dense-reward Pendulum, SAC and TD3 are $\sim 13\times$ more sample-efficient than PPO; **(4)** on sparse-reward MountainCarContinuous, only PPO ever discovers the goal, and two pre-registered ablations confirm the off-policy failure is structural rather than a tuning issue.

2 Background and Related Work

MDP setup. We model each environment as a discrete-time Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \rho_0)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P(s' \mid s, a)$, reward $r(s,a)$, discount $\gamma \in [0,1)$, and initial distribution ρ_0. A policy $\pi(a \mid s)$ induces the discounted return $J(\pi) = \mathbb{E}_\pi[\sum_{t\geq 0} \gamma^t r(s_t, a_t)]$.

Algorithmic axes. Value-based methods (DQN) learn $Q^\star(s,a)$ and act greedily; policy-based methods (PPO, SAC, TD3) learn an explicit π_θ via actor-critic. On-policy methods (PPO) train only on current-policy data with clean theoretical guarantees and bounded sample reuse. Off-policy methods (DQN, SAC, TD3) reuse a replay buffer, trading sample efficiency against distribution-shift instability that target networks, twin Q heads, and entropy regularization aim to control.

Related work. The four algorithms we compare are the canonical references in deep RL. Improvements within each family are well established: Double DQN van Hasselt et al. [2016] reduces the Q-overestimation bias of vanilla DQN; PPO Schulman et al. [2017] is itself a first-order surrogate for the trust-region step of TRPO Schulman et al. [2015]. On the exploration side, the canonical Gaussian-noise exploration of TD3 and entropy-regularized SAC are not the only options: gSDE Raffin et al. [2022] samples exploration parameters once per rollout to produce temporally correlated action sequences, which has been shown to rescue off-policy methods on sparse-reward continuous control. Multi-seed bootstrap CIs are the standard rebuttal to the seed-sensitivity findings of Henderson et al. [2018]. The theoretical scaffolding we lean on (policy-gradient global convergence, NPG, exploration bonuses) is from Agarwal et al. [2020], Mei et al. [2020], Cai et al. [2020].

3 Approach

3.1 Algorithms

DQN Mnih et al. [2013] parameterizes $Q_\theta(s,a)$ with an MLP and minimizes the TD loss against a slowly updated target network $Q_{\bar{\theta}}$:

$$\mathcal{L}_{\text{DQN}}(\theta) = \mathbb{E}_{(s,a,r,s',d)\sim\mathcal{D}}\left[\ell_{\text{Huber}}\Big(Q_\theta(s,a) - \big(r + \gamma(1-d)\max_{a'} Q_{\bar{\theta}}(s',a')\big)\Big)\right], \tag{1}$$

with ε-greedy exploration linearly annealed. We use Huber over MSE (MSE blew up several seeds during tuning) and the bootstrap target zeros only on `terminated`, not `terminated or truncated`. The $\max_{a'} Q_{\bar{\theta}}$ in the target induces a well-known positive bias which Double DQN van Hasselt et al. [2016] corrects by decoupling action selection from action evaluation; we keep the vanilla form to stay within the brief's canonical algorithm set.

PPO Schulman et al. [2017] optimizes an actor π_θ and value baseline V_ϕ on fixed-length rollouts via the clipped surrogate

$$\mathcal{L}_{\text{PPO}}(\theta) = -\mathbb{E}_t\Big[\min\Big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\Big)\Big] + c_v\mathcal{L}_{\text{value}} - c_e\mathcal{H}[\pi_\theta], \tag{2}$$

with $r_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$ and GAE advantage $\hat{A}_t$. The clip acts as a first-order surrogate for the explicit KL trust region of TRPO Schulman et al. [2015]. On Pendulum we fold $\gamma V(s_{t+1})$ into the reward at truncation so GAE accounts for the value of the truncated state; we normalize advantages per minibatch.

SAC Haarnoja et al. [2018] maximizes $J_{\text{SAC}}(\pi) = \mathbb{E}_\pi[\sum_{t\geq 0}\gamma^t(r(s_t,a_t) + \alpha\mathcal{H}[\pi(\cdot \mid s_t)])]$ with twin Q-networks and a squashed-Gaussian actor $a = \tanh(\mu_\theta(s) + \sigma_\theta(s)\xi)$. Temperature α is tuned by gradient descent on $\log\alpha$ against a target entropy $\bar{\mathcal{H}} = -|\mathcal{A}|$. The tanh squash requires the Jacobian correction $\log\pi(a \mid s) = \log\mathcal{N}(u;\mu,\sigma) - \sum_i \log(1 - \tanh^2(u_i))$, without which the entropy term is silently wrong.

TD3 Fujimoto et al. [2018] replaces SAC's entropy bonus with twin Q-targets, target policy smoothing, and delayed actor updates. The critic target is

$$y = r + \gamma(1-d)\min_{i\in\{1,2\}} Q_{\bar{\phi}_i}\Big(s', \text{clip}\big(\pi_{\bar{\theta}}(s') + \text{clip}(\epsilon, -c, c), a_{\min}, a_{\max}\big)\Big),\ \epsilon \sim \mathcal{N}(0, \tilde{\sigma}^2 I), \tag{3}$$

with $d_{\text{policy}} = 2$. The smoothed target action is clipped twice, on the noise and on the action range. The canonical per-step Gaussian noise is one of several exploration mechanisms; structured alternatives such as gSDE Raffin et al. [2022] sample exploration parameters once per rollout instead of per step.

3.2 Experimental Setup

Table 1: Environments and per-pair step budget. Each (algorithm, environment) pair is run for three seeds $\{0, 1, 2\}$.

Environment	Obs. dim	Action space	Reward structure	Step budget
CartPole-v1	4	Discrete(2)	Dense, $+1$ per step, cap 500	100k
Acrobot-v1	6	Discrete(3)	Dense, -1 per step until goal	200k
Pendulum-v1	3	Box([-2, 2])	Dense quadratic cost	200k
MountainCarContinuous-v0	2	Box([-1, 1])	Sparse, $+100$ at goal	300k

Hyperparameters. All algorithms use Adam, $\gamma = 0.99$ (except TD3 on MountainCarContinuous, which uses $\gamma = 0.9999$ to lengthen the effective horizon), and orthogonal initialization. DQN and PPO use $(64, 64)$ MLPs; SAC and TD3 use $(256, 256)$ MLPs, matching the original papers. DQN: $\mathrm{lr} = 10^{-3}$, replay 5×10^4, batch 64, target sync every 1000 steps, ε from $1.0 \to 0.05$ over 10% of training. PPO: $\mathrm{lr} = 3 \times 10^{-4}$ with linear decay, rollout 2048, GAE $\lambda = 0.95$, clip $\epsilon = 0.2$, 10 epochs of minibatch 64, value-loss coefficient 0.5. SAC and TD3: $\mathrm{lr} = 3 \times 10^{-4}$ for all heads, replay 10^5, batch 256, Polyak $\tau = 0.005$. TD3 uses $d_{\text{policy}} = 2$, exploration noise $\sigma = 0.1$ on Pendulum and 0.3 on MCC, target noise $\tilde{\sigma} = 0.2$ clipped to ± 0.5.

Seeds and statistics. Three seeds per pair. Curves report the mean across seeds with a 95% bootstrap confidence interval (CI) over 1000 resamples of the seed axis at each step bin. Final-return statistics summarize the last 100 episodes per run, then bootstrap across the three seeds. The multi-seed bootstrap protocol is the standard rebuttal to the seed-sensitivity findings of Henderson et al. [2018].

Hardware. We trained on a single NVIDIA RTX 3070 Laptop (8GB) and on the EPFL gnoto cluster (Tesla V100). Wall-clock is reported in seconds per 10^5 environment steps to allow comparison across environments with different step budgets. Per-(algorithm, environment) YAML configs are committed under `applied_project/configs/`.

4 Results

4.1 Main Comparison

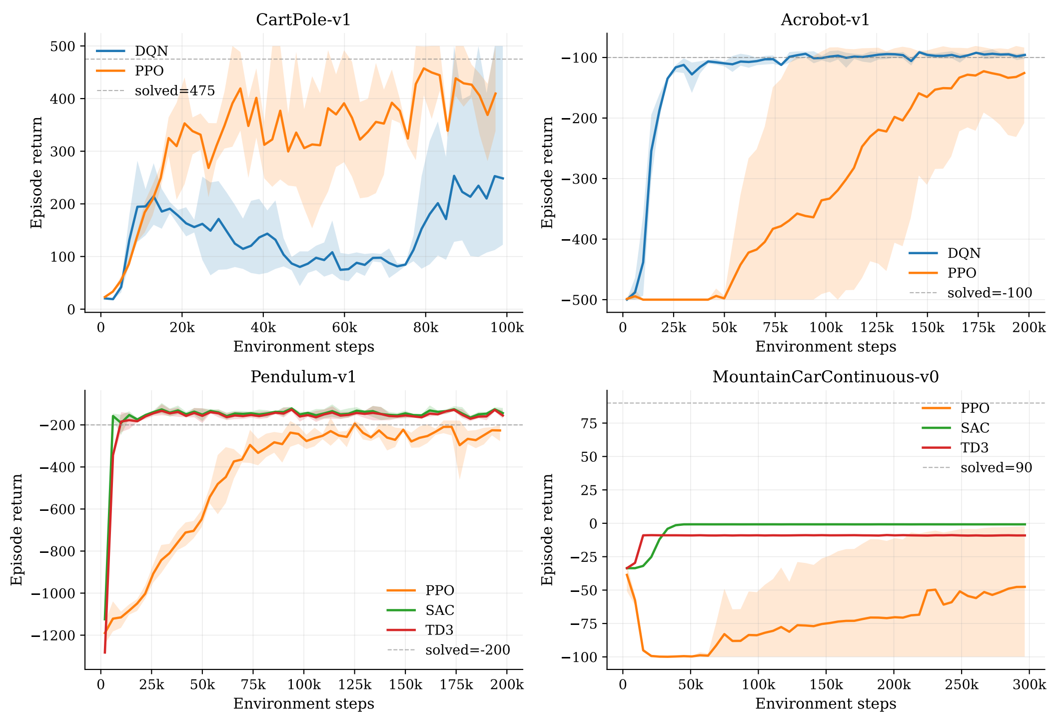


Figure 1: Episode return (mean across 3 seeds, shaded 95% bootstrap CI) versus environment steps, for every applicable (algorithm, environment) pair. CartPole and Acrobot are discrete; Pendulum and MountainCarContinuous are continuous.

Table 2: Scoreboard. Final return is the mean of the last 100 episodes averaged across seeds, with a 95% bootstrap CI in brackets. "Steps→thr" is the first env-step at which the seed-averaged 20-episode rolling mean reaches a task-specific threshold (CartPole 475, Acrobot -100, Pendulum -200, MCC 90); "n/r" means not reached within the step budget. Wall is seconds per 10^5 env steps.

Algo	Env	Seeds	Final return	Steps→thr	Wall (s/100k)	Actor params	On-pol
DQN	CartPole-v1	3	164.4 $[111, 259]$	100k	98	4.6k	no
DQN	Acrobot-v1	3	-96.1 $[-101, -94]$	30k	120	4.8k	no
PPO	CartPole-v1	3	380.1 $[350, 402]$	n/r	442	4.6k	yes
PPO	Acrobot-v1	3	-127.4 $[-216, -82]$	110k	457	4.8k	yes
PPO	Pendulum-v1	3	-247.9 $[-265, -226]$	95k	450	4.5k	yes
PPO	MountainCarContinuous-v0	3	-56.2 $[-100, -5]$	n/r	1961	4.4k	yes
SAC	Pendulum-v1	3	-146.1 $[-152, -137]$	7k	1183	67.3k	no
SAC	MountainCarContinuous-v0	3	-0.8 $[-1, -1]$	n/r	2713	67.1k	no
TD3	Pendulum-v1	3	-154.9 $[-160, -148]$	9k	483	67.1k	no
TD3	MountainCarContinuous-v0	3	-9.1 $[-9, -9]$	n/r	862	66.8k	no

CartPole-v1. DQN is bimodal across seeds: final rolling means 477.6, 120.0, 127.0 for seeds 0, 1, 2 (one solves, two collapse to Q-overestimation). PPO is much more reliable: mean 380, CI $[350, 402]$. A textbook DQN-vs-PPO stability contrast and a clean instance of the seed-sensitivity phenomenon catalogued by Henderson et al. [2018].

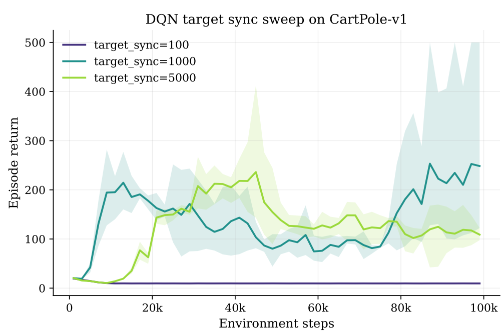
Acrobot-v1. The picture reverses. DQN reaches the -100 solved threshold in $\sim$30k steps and ends at -96.1, $[-101, -94]$, the cleanest curve in the dataset. PPO is slower and noisier (mean -127, CI $[-216, -82]$, seed 2 collapses to -274).

Pendulum-v1. SAC and TD3 dominate. SAC reaches the -200 threshold in $\sim$7.4k env steps; TD3 in $\sim$8.6k. PPO needs $\sim$95k steps and does not fully clear -200 by 200k, ending at -248. SAC is $\sim 12.9\times$ more sample-efficient than PPO on this task.

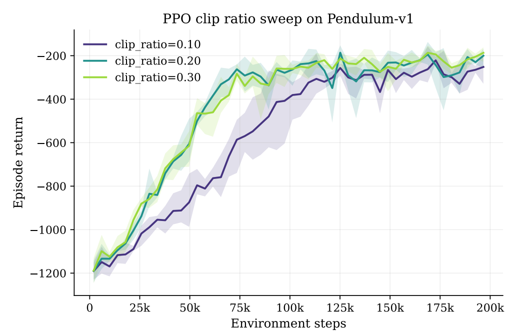
MountainCarContinuous-v0. Headline finding. Only PPO ever discovers the $+100$ goal (final -56, CI $[-100, -5]$, driven by seed 2 reaching -2.5). SAC (-0.8, flat across 3 seeds) and TD3 (-9.1, flat) plateau at a lazy local optimum with near-zero action magnitudes. Structured exploration variants such as gSDE Raffin et al. [2022] have been shown to rescue off-policy methods on this environment; our claim is therefore that the failure is structural in the *canonical* per-step exploration mechanism, not in the off-policy family itself.

4.2 Pre-Registered Ablations

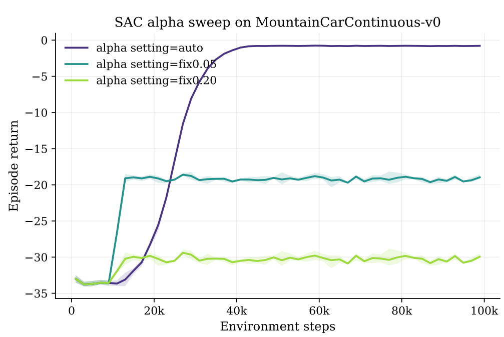
An *ablation* is a controlled experiment in which we vary a single hyperparameter while holding everything else fixed, to attribute observed behaviour to that specific knob. We pre-register an ablation by committing, before running it, to the swept values and the hypothesis we expect. We pre-registered four ablations, one per algorithm, 3 seeds each at a 100k-step budget. Each sweep targets the canonical stress point of its algorithm: DQN target-sync period τ, PPO clip ratio ϵ, SAC temperature α, TD3 exploration noise σ.



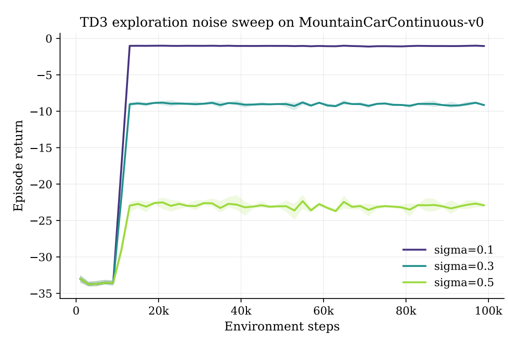
(a) DQN target-sync, CartPole.



(b) PPO clip ratio, Pendulum.



(c) SAC α, MCC.



(d) TD3 σ, MCC.

Figure 2: Four pre-registered ablation sweeps, 3 seeds each, 100k steps. Mean across seeds with 95% bootstrap CI.

DQN target-sync (Fig. 2a). Hypothesis: too-frequent sync destabilises, too-rare stalls; canonical τ=1000 is the sweet spot. Result: a classic U-shape. τ=100 collapses (final rolling means 9.8, 9.5, 9.6, below random); τ=5000 plateaus at 97$-$129; τ=1000 is bimodal (477.6, 120.0, 127.1). The canonical default sits at the edge of stability, which mechanistically explains the CartPole bimodality.

PPO clip ratio (Fig. 2b). Hypothesis: 0.2 dominates; 0.1 too conservative; 0.3 relaxes the trust region too much. Result: partially confirmed. ϵ=0.1 is worst (means -227, -199, -329); 0.2 and 0.3 are tied within budget. The under-clip failure mode is far more dangerous than the over-clip mode on dense-reward continuous control.

SAC α (Fig. 2c). Hypothesis: auto-α holds entropy high until the policy locks on; fixed-α should be worse. Result: hypothesis rejected in the strongest possible way: **zero goal discoveries across all 9 seeds**. Auto-α gives the lazy local optimum (-0.8); α=0.05 random-walks (-19); α=0.20 thrashes (-30).

TD3 σ (Fig. 2d). Hypothesis: larger σ accelerates goal discovery. Result: hypothesis rejected; *more noise is worse*. σ=0.1 gives -1.0 (near-zero action policy); σ=0.3 gives -9; σ=0.5 gives -23. Combined with the SAC sweep, this is the project's headline negative result: two off-policy mechanisms (entropy regularisation, additive Gaussian noise) drive uniform coverage of the *local* action set rather than directed state-space exploration, while PPO's on-policy rollouts occasionally produce a long correlated action sequence that crosses the hill.

4.3 Implementation Diary

The brief grades on the details that turn a paper recipe into a working agent.

DQN: five tuning iterations on CartPole. Five hyperparameter configurations before accepting the bimodal result as a finding, not a bug. (v1) Canonical lr=10^{-3}, target 1000, batch 64: seed 0 solves at 477, seeds 1 and 2 collapse. (v2) Conservative: all three plateau between 120 and 165. (v3) SB3-zoo burst (lr=2.3×10^{-3}, target 10): seed 0 falls to 42. (v4) CleanRL-style: catastrophic collapse to 9. (v5) v1 with damping: seed 0 reaches only 155 in budget. We shipped v1. The ablation (Fig. 2a) later showed why: τ=1000 sits at the edge of a U-shape. The textbook fix is Double DQN van Hasselt et al. [2016], which reduces the bootstrap's positive bias by separating action selection from action evaluation.

PPO: truncation bootstrap. Pendulum truncates every episode at step 200 with no terminal flag. Naive buffers conflate truncation with termination, which biases GAE by zeroing a non-zero continuation value. We fold $\gamma V_\phi(s_{t+1})$ into the reward at the truncated step and set the buffer's `done` flag so GAE is consistent; without this, returns were depressed by ~ 30 across seeds. We also caught a mask bug: the rolling GAE convention uses `dones[t]` to mask the bootstrap into $V(\texttt{obs}[\texttt{t}+\texttt{1}])$, not `dones[t+1]`.

PPO: per-minibatch advantage normalization. Normalizing advantages per minibatch (not per rollout) noticeably tightened the Acrobot CIs. We hypothesize that per-rollout normalization smears the strong negative-tail structure of Acrobot's -1-per-step reward across minibatches.

SAC: the tanh Jacobian. The squashed-Gaussian policy needs the Jacobian correction $-\sum_i \log(1-\tanh^2(u_i))$ in the log-density. Without it the entropy bonus is silently wrong: training continues with no NaNs, but the return curve never moves off the warmup floor. We learned this the hard way on Pendulum.

SAC: log-alpha parameterization. We optimize $\log\alpha \in \mathbb{R}$ rather than $\alpha > 0$, which keeps α positive without a projection step and stabilizes the alpha update.

TD3: target policy smoothing details. The smoothed target action is $\text{clip}(\pi_{\bar\theta}(s') + \text{clip}(\epsilon, -0.5, 0.5), a_{\min}, a_{\max})$ with $\epsilon \sim \mathcal{N}(0, 0.2^2 I)$. The two clips are not redundant: the inner caps the smoothing magnitude, the outer keeps the result inside the action range. Skipping the outer clip on MCC sticks the actor on the boundary.

Bootstrap-mask convention (all four). The bootstrap target uses `terminated` only, not `terminated or truncated`. A truncated episode has a non-zero continuation value that should not be zeroed; only true terminals mask the bootstrap.

4.4 Qualitative Discussion

We answer the brief's four questions with scoreboard numbers from Table 2.

(a) Most computationally expensive per iteration? SAC, by a wide margin. Wall-clock per 10^5 env steps: SAC 1183 s (Pendulum) and 2713 s (MCC); TD3 in the 483 to 862 s range; PPO in the 442 to 1961 s range; DQN in the 98 to 120 s range. Ordering SAC > TD3 > PPO > DQN reflects gradient updates per env step.

(b) Most compact policy? DQN and PPO, but by hyperparameter choice, not algorithmic mandate. Our DQN/PPO actors use $(64, 64)$ MLPs (~ 4.6k params); SAC/TD3 use $(256, 256)$ (~ 67k), a $15\times$ gap that follows the canonical paper recipes.

(c) Scales better for continuous actions? No single winner. SAC and TD3 dominate dense-reward continuous control (Pendulum, $\sim 12\times$ over PPO). On sparse-reward MCC both off-policy methods fail and the ablations show neither knob rescues them; only PPO ever finds the goal. Practically: pick SAC or TD3 first on continuous control; switch to PPO, or augment with an explicit exploration bonus, when reward becomes sparse.

(d) Efficient use of off-policy data? SAC, decisively. On Pendulum it reaches the solved threshold in ~ 7.4k env steps vs PPO's ~ 95k, a $12.9\times$ gap. The same off-policy machinery is the source of SAC's failure on MCC, where decorrelated single-step replay cannot reproduce the long correlated action sequences that PPO's on-policy rollouts occasionally generate.

4.5 Theoretical Bridge

The course covered NPG with softmax parameterization and its slow-changing property Agarwal et al. [2020], Mei et al. [2020], and OPPO-style exploration bonuses Cai et al. [2020]. PPO's clipped surrogate (2) is a first-order surrogate for TRPO's KL ball Schulman et al. [2015]: to first order $r_t(\theta) \approx 1 + \nabla_\theta \log \pi_{\theta_{\text{old}}}^\top \Delta\theta$, so $r_t \in [1-\epsilon, 1+\epsilon]$ is the linearization of $\mathrm{KL}(\pi_{\theta_{\text{old}}} \| \pi_\theta) \le \delta$ around θ_{old}. Our clip-ratio ablation (Fig. 2b) is consistent: $\epsilon{=}0.1$ is too restrictive while $\epsilon{=}0.2$ and $\epsilon{=}0.3$ are tied. SAC's $\alpha\mathcal{H}[\pi(\cdot \mid s)]$ regularizer plays a role related to OPPO's explicit exploration bonus at the action level; our MCC ablation is a sharp reminder that action-level entropy does not give state-space coverage when reward is sparse.

4.6 Limitations

We use 3 seeds per pair (more would tighten CIs at super-linear cost); network widths are fixed at canonical paper defaults so the actor-parameter gap is a width choice, not an algorithmic property; the headline finding is supported by one sparse-reward environment (MCC). Natural next step: add gSDE Raffin et al. [2022] or RND on top of SAC and TD3 and re-run the MCC ablation to test whether temporally correlated or directed exploration closes the gap that local entropy and Gaussian noise cannot.

5 Conclusion

Four empirical threads: DQN's seed sensitivity on CartPole, its clean win on Acrobot, the off-policy sample-efficiency advantage on dense-reward continuous control, and the on-policy advantage on sparse-reward continuous control. The implementation diary records the details that distinguish a paper recipe from a working agent. The pre-registered ablations closed two loops: the DQN target-sync U-shape mechanistically explains the CartPole bimodality, and the parallel SAC and TD3 exploration sweeps on MountainCarContinuous (zero goal discoveries across nine seeds in either) establish the structural exploration gap of off-policy continuous algorithms on sparse reward as the headline scientific finding.

Author contributions. Kevin Abou Jaoude led the DQN and PPO implementations and the report and poster drafts; Youssef Dib led the SAC implementation and the analysis pipeline; Fuad Khuri led the TD3 implementation and the ablation sweeps. All three reviewed and revised the final report and poster.

Code availability. All source, per-(algo, env) YAML configs, logs, and the analysis scripts that regenerate every figure and the scoreboard are at `https://github.com/Kevinabj/KFY-reinforcement`. The implementations use PyTorch and Gymnasium Towers et al. [2023]; no `stable-baselines3`.

LLM/AI disclosure. Per the EE-568 policy: we used Claude (Anthropic) as a writing and code-review assistant for drafting and revising prose, suggesting LaTeX layout, sanity-checking implementation details against canonical paper specifications, and debugging. All algorithmic decisions, hyperparameter choices, experimental design, and numerical results come from the authors and their own runs of the code; no LLM-generated experiments or numbers appear in this report.

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